Failure Modes

Spot it, name it, find the lever that fixes it

You found something wrong in a trace. Find the card that matches, then pull its lever. Each mode is named for **the lever that fixes it**.

1 · Tool

The agent's connection to the outside world breaks down.

- Wrong tool chosen
- Right tool, wrong arguments
- Never called when it should have been
- Output misread
- Tool errored; agent carried on

Look in: tool spans: inputs and outputs.

Lever: tool names, descriptions, input/output schemas.

2 · Grounding

It asserts things nothing it retrieved supports.

- Fabricated facts, IDs, numbers
- Nothing retrieved, answered anyway
- Contradicts the right document
- Accepts a false premise

Look in: final answer vs. tool and retriever outputs.

Lever: retrieval and knowledge: what's retrieved, whether ungrounded answers are allowed.

3 · State

What it needs is lost, stale, or poisoned.

- Context truncated or degraded
- Bad intermediate written down as truth
- Forgets across turns

Look in: the thread; late spans vs. early inputs.

Lever: memory design: what persists, where, what enters the window.

4 · Control

The path is wrong: doesn't stop, doesn't finish, or wanders.

- Infinite loop
- Budget drain
- Stops too early
- Bad or abandoned plan

Look in: the trace's shape: count, order, tokens, time.

Lever: scaffolding: control flow, routing, step limits.

5 · Specification

Did what it was told, which wasn't what was meant.

- Hidden assumption never stated
- Instructions fade over a long run
- Task quietly redefines itself
- Satisfies the metric, defeats the purpose

Look in: system prompt vs. final answer.

Lever: instructions: system prompt, success criteria, output format.

6 · Safety

Harmful or disallowed, even if it completes the task.

- Acts without authorization
- Prompt injection
- Leaks credentials, PII, policy
- Appears to comply, doesn't

Look in: actions taken; external content in tool outputs.

Lever: guardrails, permissions, sandboxing, human-in-the-loop.

Describe it

Dimension	Ask	Trains
Silent ↔ loud	Would anyone notice without the trace?	Detection
Isolated ↔ cascading	Where is the *earliest* bad span?	Localization
Model ↔ code	Probabilistic, or a deterministic bug?	Attribution

Walk the trace

Ask these in order. Stop at the first **yes**: that's your mode and your lever. Then keep going, because traces often hold more than one.

1. **Did it do something harmful or unapproved?** Acted without permission, leaked data, obeyed instructions hidden in a tool's output. → Safety pull **guardrails, permissions, human-in-the-loop**
2. **Is a tool span wrong?** Error ignored, wrong tool, wrong arguments, misread output, or a tool that should have been called and wasn't. → Tool pull **tool names, descriptions, schemas**
3. **Is the shape wrong?** Repeats, runs huge, stops short, wanders. → Control pull **control flow, step limits, routing**
4. **Does the answer say something no tool returned, or contradict one?** → Grounding pull **retrieval and what the prompt allows**
5. **Did it lose, forget, or misuse something from earlier?** → State pull **memory design**
6. **Did it do what was said, but not what was meant?** → Specification pull **instructions and success criteria**
7. **None of these?** It may be outside the six: wrong math, wrong tone or audience, policy broken, over-refusal, guessed instead of asking, unequal treatment, too slow, inconsistent. Or it isn't a failure *for this agent's purpose.* → Not in the six name it anyway and say who it hurts